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Bridging the Gap: Enhancing Voter Confidence in EVMs through Stream lit-powered Interfaces

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***Abstract*— The use of EVM and its accuracy and speed have changed the election procedures in India. But the question of impersonation in casting votes is also a cause for worry and distrust among EVM-skeptics, especially from rural or left-wing extremism (LWE)-affected areas. The research looks into integrating the Streamlit framework for EVMs to enhance interfaces that will increase voters' trust and make them visible and engaging. Using Streamlit' ability to make the Python code interactive, we suggest a novel solution that makes it easier to interact with EVM interfaces that provide real-time feedback and encourage voter verification via Aadhaar and phone number linkage. After the election, our system verifies commitments to voters by sending verification messages. Adam encryption takes a user-centric design and iterative development approach to solve the complex electoral technology-voter impersonation dilemma in order to foster transparency and trust in our democracy.**

Keywords— Electronic Voting Machines(EVMs), Streamlit framework, Voter transparency, Voter engagement, Electoral integrity, Voter trust, User acceptance, Security vulnerabilities, Data protection, Chief Election Commissioner.

I. Introduction

In the past, democratic elections were held using ballots, which were pieces of paper that had been written on by the voters. However, today in India, EVMs, or electronic voting machines, are used. This has modernized the voting process but has also led to serious vote reliability concerns. While EVMs are designed to secure votes, if physical access is compromised, it could open them up to tampering that can put the entire election system at risk. A quick vote could go a long way toward determining the fate of the country.

The problem of voter impersonation poses an acute challenge. The campaign team of a few parties often labors to push the people, albeit sometimes forcefully, to cast their vote in the name of the candidate on multiple identity cards. It becomes an exercise that gets going without the much-needed will or knowledge of the actual voter. The menace is prevalent in rural areas and Maoist belt regions. A large number of voters are, as a result, unable to prove that they cast their vote. So, they end up not being able to fight electoral malpractices only because they cannot prove that they voted.

As a solution to this issue, this paper suggests a new technique that uses Python Streamlit. There will be two interfaces on this system: the Chief Election Commissioner and the Election Poll Worker can log in using the second one during the election period. Once voting takes place, poll workers will record every vote in a machine (existing voting machine provisioning features) and then forward the vote for the respective constituency to the Chief Election Commissioner. Before voting begins, every elector will receive a notification from the Election Commission that he is a verified voter; for an unverified vote, the elector shall respond to the text message to approve his identity as a verified voter.

With our strategy, vote impersonation will be averted because it provides a democratic way to validate votes. We identify and immediately stop fraudulent activity by allowing voters to authenticate their participation. With the introduction of this technique, the privacy of voters' responses will be ensured, as well as enhanced clarity and security for the process. This measure will secure the integrity of Indian elections and will include this approach, thereby safeguarding both democracy and the certainty of precise vote counting for all votes cast.

II. Related work

[1] The study "Blockchain-Based E-Voting and Electoral Fraud Detection" by Prof. Sumit Shevtekar and Varad Kalambarkar (2023) made use of blockchain technology, smart contracts, cryptography, and privacy-preserving techniques. According to their research, blockchain makes votes time-stamped, unchangeable, and impenetrable, which improves e-voting security and makes real-time fraud detection easier. They stressed the significance of safe voter registration and identity verification. The report did, however, highlight the necessity of strong regulatory frameworks, possible problems with government database interoperability, and continued security research to deal with new threats.

[2] In 2021, Jongsoo Sohn, Kyung Eun Kim, Soohan Park, and Hyunyeon Kim unveiled the "E-voting System Using Blockchain Technology and Homomorphic Encryption to Encrypt Voter Data." The study showed that voters can validate their ballots in plain text and conduct in-depth voting

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analysis by using encrypted voter data instead of the votes themselves, which maintains voter anonymity. This method expedites counting and boosts efficiency. However, there are drawbacks, such as possible speed problems when homomorphic encryption is used extensively and worries regarding data security.

[3 In 2024, Natarajan et al. created a blockchain-integrated online voting system that used the AES algorithm with OTP validation. Their study shed light on issues like third-party certification and greenwashing by examining the interactions between blockchain technology, cryptography, and agri-food supply networks. They listed the glaring lack of financial incentives for farmers as one of the primary barriers

[4] Blockchain's potential to improve data security and transparency was examined in "A Survey on Blockchain for Information Systems Management and Security," a study done in 2021 by Berdik et al. According to the study, voter registration and voting system security are enhanced by blockchain's decentralization. But it also pointed out possible negatives, like network vulnerabilities, insider threats, and accessibility problems, emphasizing the need for improved design and awareness.

[5] Adeniyi et al. (2024) presented a private/public key cryptography created by biometrics for a blockchain-based electronic voting system. They used the elliptic curve digital signature algorithm, biometric data (fingerprints), and blockchain technology. The integration of blockchain, biometrics, and cryptography in electronic voting—which prioritizes private key generation—is one of the major discoveries. Limitations, however, include worries about the security, privacy, and dependability of electronic voting systems.

[6] Annadate et al. (2017) introduced an online voting system that combines biometric verification and ZigBee technology with the aim of improving the voting process. With an emphasis on the automation of voting processes, they conducted a comprehensive analysis of voting systems, biometrics, and electronic voting machines (EVMs). However, the results of their analysis pointed to significant shortcomings, including issues with implementation, security, and scalability.

[7] In his proposal for a "Blockchain-based e-voting system in a university," Adil Marouan (2024) combined blockchain technology with ring signing, smart contracts, digital signatures, and voter anonymity. According to the study, this approach offers a significant improvement in execution time and scalability along with benefits in transparency, anonymity, and self-tallying. Voter integrity and validity are guaranteed by advanced cryptography. To increase overall efficiency, challenges include cutting gas prices, controlling transaction latency, and interfacing with current platforms.

[8] The Technology Acceptance Model and surveys were utilized by Kamoliddin Murodjon ugli Mannonov and Seunghwan Myeong in their 2024 study, "Citizens' Perception of Blockchain-Based E-Voting Systems: Focusing on TAM," to show that there is a favorable correlation between citizen attitudes and the adoption of blockchain e-voting. Trust and usability were highly dependent on perceived security. The theoretical character of the study and the requirement for actual research on internet access and infrastructure are among its limitations.

[9] In 2018, Mrs. Malati V. Tribhuwan, Mrs. Harsha V. Patil, and Mrs. Kanchan G. Rathi carried out "A Study on Decentralized E-Voting System Using Blockchain Technology." Blockchain technology, hashing for integrity and anonymity, and decentralized applications were all used in the study. Important conclusions showed how blockchain technology might improve voting process security and integrity by ensuring voter anonymity through the use of hashed tokens. In order to guard against fraud and manipulation, the study stressed the importance of having a transparent and safe voting mechanism. However, it was also mentioned that more study and development were needed because the system was still in the conceptual stage. There are issues with how well blockchain solutions scale for high vote counts, and successful blockchain adoption requires public acceptance and comprehension of the technology.

[10] Marky et al. (2024) examined voters' perceptions of printed physical audit trails for online voting using capacitive sensors and QR codes. The gathering and analysis of data in online hybrid voting systems was the primary focus of the study. The results emphasized issues with voter trust, usability, and security. Among the drawbacks were possible participant bias and inadequate processing of qualitative data.

[11] An online voting system with a centralized design, user authentication processes, and homomorphic encryption for security was proposed by Ahmed and Pattanasetty (2024). TCP, characteristics, and use case diagrams were crucial components. Security, privacy, scalability, and performance concerns were identified; nonetheless, there were drawbacks, including a lack of information about the implementation.

[12] An Enhanced Implementation of Aadhar-Based Electronic Voting Machine," created in 2019 by Naganaboina Krishnaveni, Dr. Yogesh Kumar Sharma, and Dr. S. Nagaprasad, uses GSM for OTP, RFID, and Aadhar-based biometric authentication. By using biometric verification to confirm voters, the system reduces fraud and increases security and efficiency. Infrastructure requirements in rural areas and reliance on the veracity of the Aadhar database present challenges.

[13] Voter Token Security (VTS), digital signatures, smart cards, and audit logs were used in the development of an online electronic voting system for Kosovo in 2012 by Dr. Blerim Rexha, M. Sc. Vehbi Neziri, and M. Sc. Ramadan Dervishi. Although client-side security, ongoing monitoring,

and auditing needs present obstacles, the system increases voting flexibility, security, and possible cost savings.

[14] The "Design of Distributed Voting Systems" thesis by Christian Meter from 2015 examines electronic voting systems in Australia, Norway, and Estonia, pointing out security flaws and suggesting a safe e-voting template. The study emphasizes improved security using blockchain technology and cryptographic approaches with a modified proof-of-stake mechanism, but it also points out issues with voter coercion, verifiability, operational security, and scalability.

III. METHODOLOY

An Overview of Streamlit

Streamlit is an open-source Python package for creating custom web apps that require no, or very little, programming effort. With this one, developers can also build data-driven apps and deploy them quickly in a responsible way.

Benefits of Streamlit:

Ease: Use Streamlit's simple API to create sophisticated apps with a few lines of pure Python; no knowledge of HTML, CSS, or JavaScript is required!

• Quick Development: Its framework comes to the rescue by allowing changes in real-time after updating your code. This makes it easy to prototype your solution.

• Common Python Libraries Integration: Streamlit easily integrates with NumPy, which is used for numerical computations, Pandas (data processing), and Matplotlib (including its extension Bokeh), which are popular choices among data visualization packages.

• Interactive Components: The design even has several built-in interactive components like text entry boxes, buttons, and programmers to boost user interfaces.

Streamlit makes deployment easy. Your Streamlit app could be deployed on a number of platforms, which helps in its easy distribution and access such as Heroku, AWS, and Streamlit Sharing.

Data Collection and Integration

In Figure 1 Linking and Storing Voter Information: Voter ID, Aadhaar, and mobile numbers are among the major data sources used by the system. These sources are made available and maintained securely. This allows the use of a centralized SQLite database that can integrate these data points to reflect the unique identity of an individual to its respective voter. It reduces the chances of impersonation by correctly managing the voter's data, ensuring it is not shared with any unauthorized party, and ensuring checks are performed.

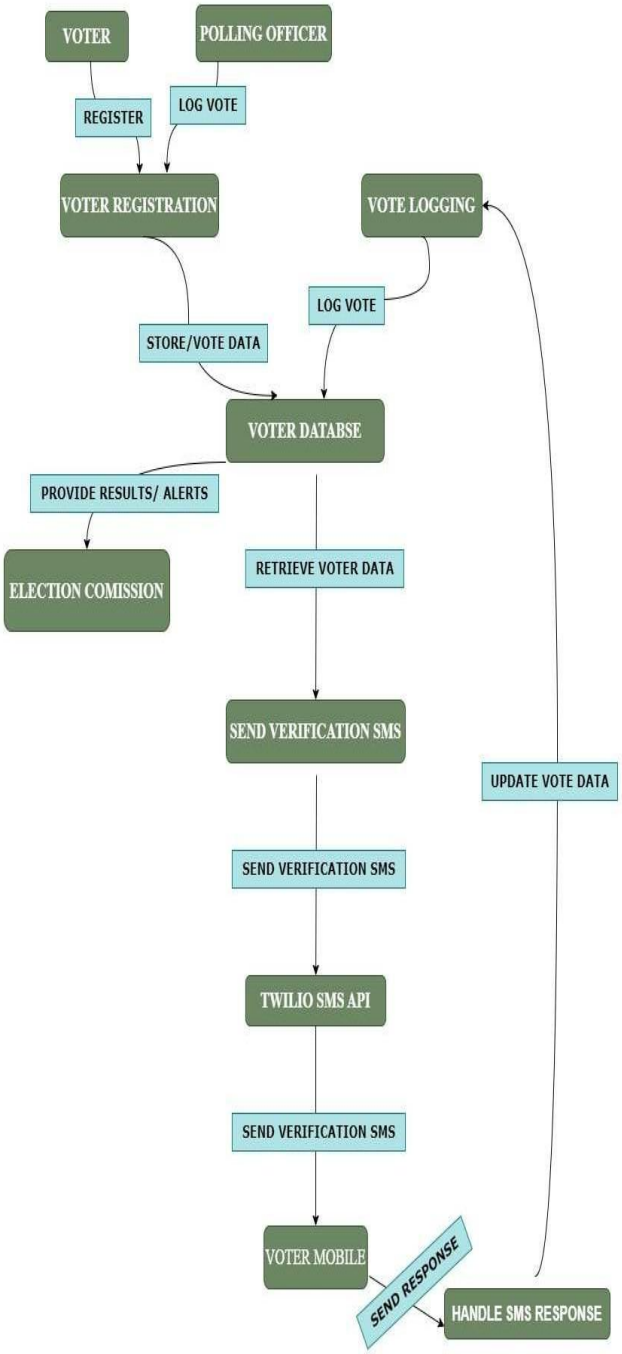

Figure 1. Data Flow Diagram (DFD)

Data Creation And Structure

The voter data is stored in an SQLite database, considering it a good choice due to its high speed and capability of boasting. One table, Voter, consists of a database structure. The voting status of voters may be efficiently queried and updated with the help of this schema model on the test data. A model has now an opportunity to utilize a number of algorithms and predict models' predictions on different images.

System Architecture

Streamlit-Based Interfaces: In Figure 2 The interface allows voter data to be input and registered. It comes with the options to search for voters and view voter lists, as well as insert voter ID, Aadhaar number, mobile number, and name.

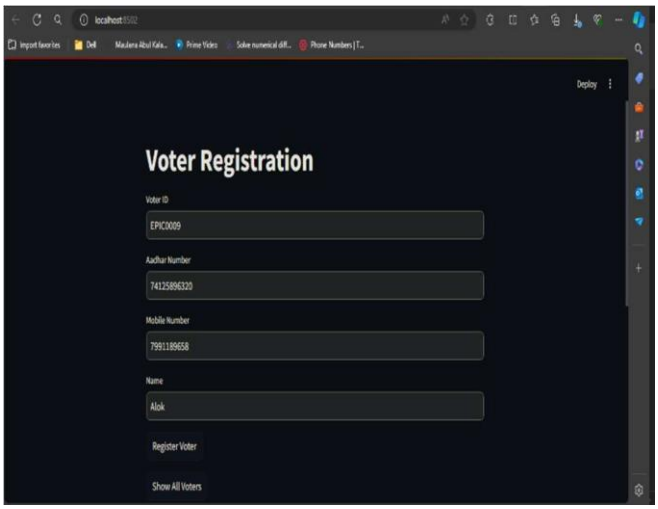

Figure 2. Voter Registration Interface

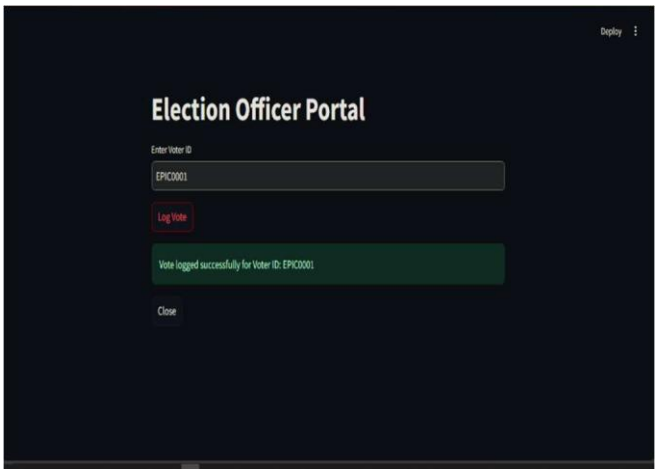

Figure 3. Polling Officer Portal

In Figure 3 Polling Officer Portal: This portal is used for vote logging and is meant only for polling officers. When poll workers type in a voter's ID, the system updates itself to note that this person has voted.

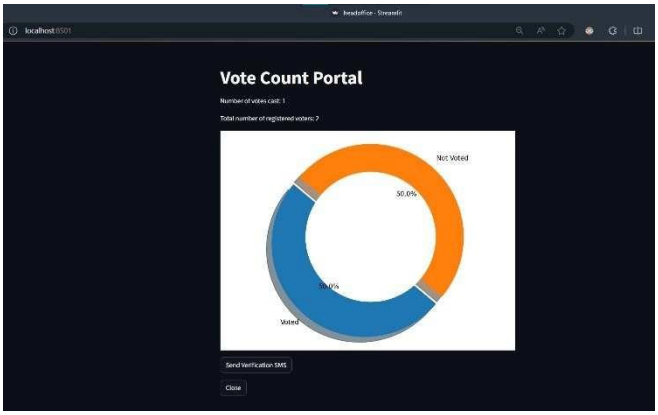

Figure 4. Voter Record and Communication

Voter Record and Communication: The Polling Officer Portal casts and records votes in order to update the voter's database status. Every piece of information is sent to the Election Commission's database so that it can be processed and counted centrally. Updates can be reliably and accurately transmitted back to the central system in this manner.

Verification Process

Sending Verification SMS: SMS notifications are delivered to all those users who cast their vote after the election. In case the same person doesn't cast their vote, the communications are sent to let them know that they've already put in their votes and would only need a reply. Here's what the SMS entails: Your vote has been confirmed, Dear [Voter Name/Balloter]. If you haven't done so already , please replywith a 1.

Response Mechanism: The system is designed to handle responses from voters who claim they did not vote. Responses are collected and analyzed to detect any discrepancies. If a voter replies with "1," indicating they did not cast their vote, this information is flagged for further investigation to identify potential cases of voter impersonation .

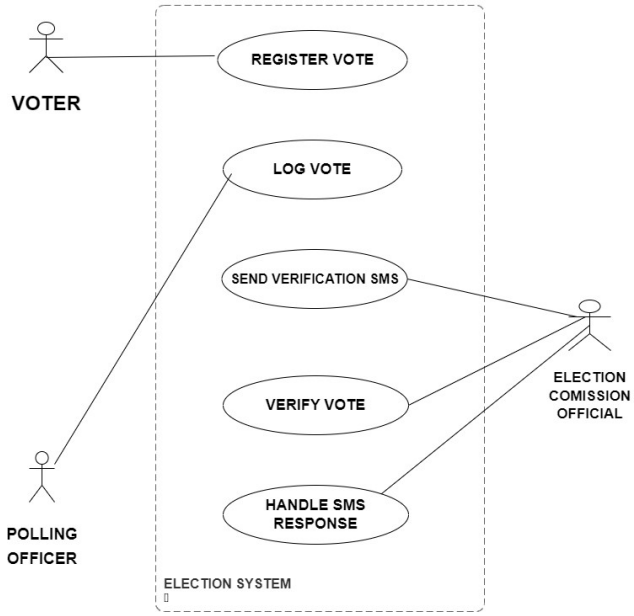

Figure 5. Use Case Diagram (UCD)

Security Measures

IN Figure 6 Data Protection: By using encryption techniques, voter personal data, including phone numbers and Aadhaar numbers, is assured to remain secure. Restriction on system access.

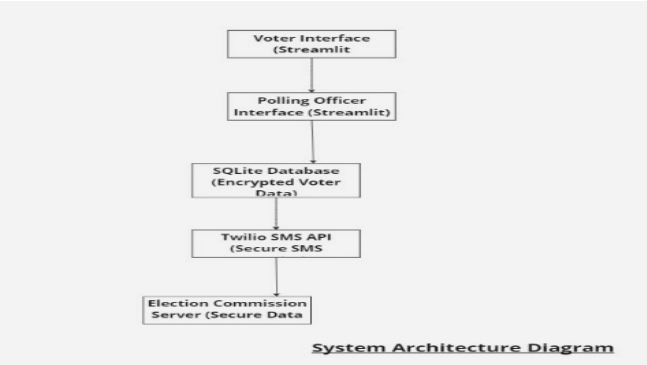

Figure 6. System Architecture Diagram (SAD)

Secure Communication: All SMS communications are processed securely by Twilio. Twilio Platform: messages are securely sent (encrypted) so that third parties cannot read them without permission. To avoid unwanted use, Twilio credentials—which include the authentication token and account SID—are securely maintained and not hardcoded in the application code

IV. Implementation

A. Overview

In this system, you would create the web application using Streamlit, a SQLite database to store data safely, and implement SMS alerts through the Twilio API. The idea is that, by way of a very rigorous verification process, you can add to voter trustworthiness and reduce the chance of your own appearance in voting.

B. Setting up an environment

To configure the development environment, the following steps were performed:

Software prerequisites:

- Python 3.8 or above: It provides fundamental structures and versatile libraries and is selected because it is easy to learn.
- Streamlit is an efficient choice for interactive web app creation.
- SQLite is widely recognized for its straightforward, efficient database management.
- Twilio is reputed for its reliable SMS messaging services.

Setup process:

- Install Python and the Python libraries (pip: The official package manager for Python was used for installation.
- Secure records of the details of people who voted—this STEP was implemented using SQLite.

C. Database Structure

A database should securely store voter ID, Aadhar number, cellphone number, name, and voting status. Vital fields of the voter entity include the following:

- ID: primary key, an integer that is unique to every voter record. voter_id: It is the voter's ID text field.
- aadhar_number: Here you will enter the voter's Aadhaar number in this text field.
- name: The voter's name, entered as a text field.
- mobile_number: The voter's mobile number entered in text field.
- Voted: a boolean field that indicates whether the voter has voted.

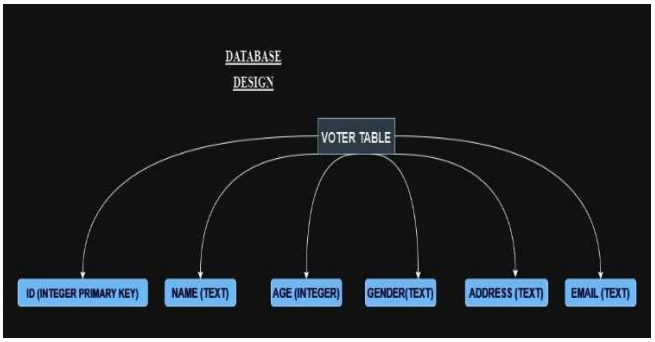

Figure 7. Database Design

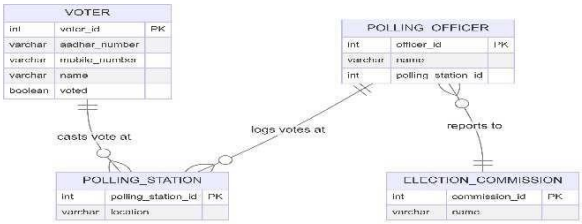

Figure 8. entity-relationship (ER) diagram.

In Figure 8 The entity-relationship (ER) diagram. database schema was drawn as an

D. Streamlit Applications

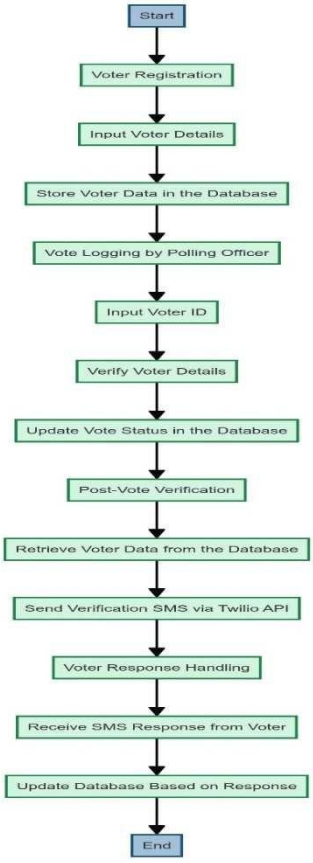

Figure 9. **Stream lit Application Flow Diagram**

In Figure 9 Multiple interfaces are included in the Streamlit program to make voting, vote tracking, and verification easier

1. Voter Registration

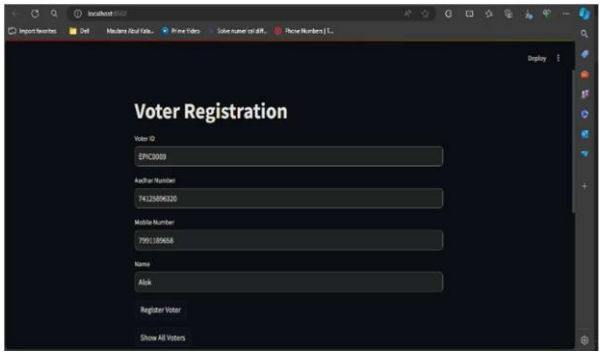

Figure 10. Voter Registration Interface

In Figure 10 The information from users via the voter registration interface Store safely in SQLite DB. The interface has spaces for the voter ID, the name of a citizen, his mobile number, and his Aadhaar number. The input is checked; a confirmation message is shown, and the user accepts it, from which point an entry is saved.

2. Vote Count Portal

The votes are logged using the vote-logging interface for poll workers. Poll workers enter a voter's ID, and the system queries against records to find out if it is allowed for voting or not, while after votes have been cast, tables are updated with vote information. This way, every vote is counted, and the voter has not yet voted.

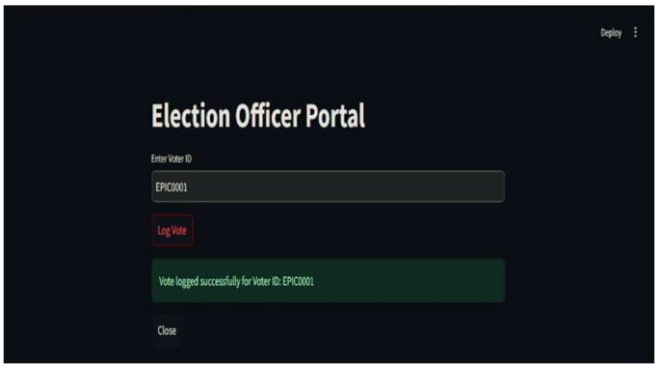

Figure 11. Polling Officer Portal

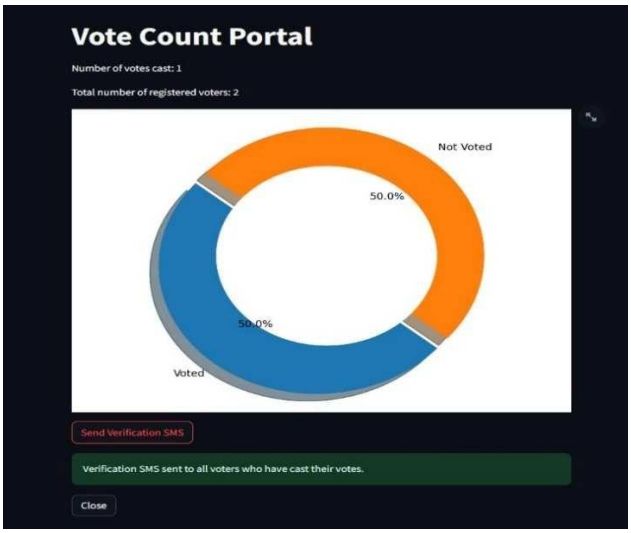

Figure 12. **Election Commission Portal**

E. Verification Process

In Figure 13 we can see that SMS alerts are sent to voters after they have voted, which is part of the verification processthat maintains sanctity in elections. The Twilio API has been sending these notifications, essentially asking if you are really a voter who voted.

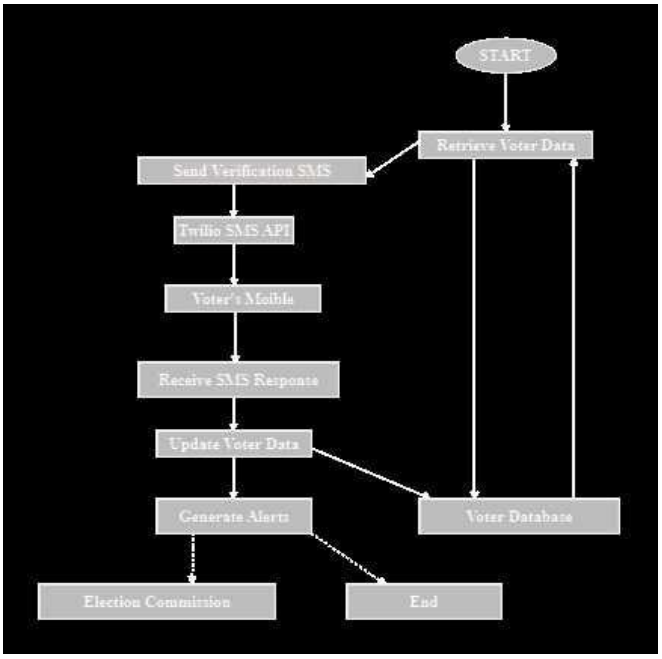

Figure 13. **SMS Verification Flowchart**

A. Sending Verification SMS:
The system in turn receives voter information from a related database and sends SMS to those who have already cast their votes. If the voter did not vote, it tells you what to do, and if they did, there is information on how that ballot was cast.

B. Handling Responses:
With the SMS notifications we provide, voters then simply respond to them, and their responses are processed in order for us to update our database with votes. If the voter says they did not vote, a circuit will alert the election commission and flag that individual for follow-up investigation

F. Security Measures

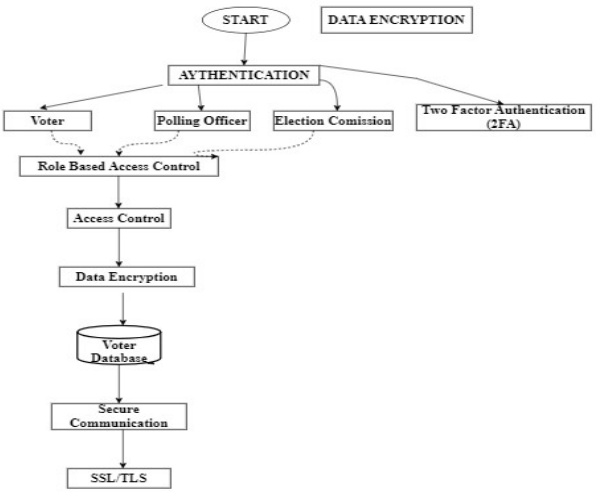

Figure 14. Security Measures Diagram

In Figure 14 Security protocols are critical to protecting voter information and ensuring the integrity of the system. The following measures were implemented:

- Security for sensitive voter data Suitable encryption: Sensitive voter data is protected both in transit and at rest.
- Communication over SSL: To prevent interception, all communication with the Twilio API takes place using HTTPS.

only access control: authorized persons can only use the database and API credentials

G. Testing and Validation

Comprehensive testing and validation were conducted to ensure the reliability and accuracy of the system.

- Unit Testing: We have checked every component to see if it works fine alone.
- Integration testing We tested from bottom to top to get every component of the system working as a whole.

To evaluate the system from an end-user perspective, real-world scenarios were simulated using user acceptance testing

H. Deployment

Deploying the application involved several key aspects:
Web App Running on Streamlit: This is where the web application was prepared to work on the Streamlit server only.Database Scaffold: Initializes the necessary schema andinitializes the SQLite database.
Twilio Integration: Securely managed Twillio API credentials and configured the Twilio API for sending SMS notifications.
Then, the program was deployed on a server so that it could still be accessed by users while maintaining very high security and dependability standards.

I. Future Enhancements

Potential improvements and additional features were identified for future iterations of the system:

- A Well-Built Database System: Employing a robust database system that is also more scalable to deal with large chunks of data.
- Increased security measures: added another layer of security, including two-factor authentication.
- Biometric Verification: Unauthorized voting is largely due to inaccurate voter verifications, and hence accuracy can be improved by adding ways for biometric identification.

V. Result And Discussion

A. Testing

We ran the app through extensive manual testing to ensure it was reliable, safe, and functional. We tested the following components:

- Voter Registration The status of new voter registration against Voter Registered in Supplement was cross-verified and verified that by entering either one option from the a) (Voter ID/Aadhar Number/Mobile Number/Name), it requires to allow on clicking the 'Search' button. The test cases certainly worked, and the voter data was properly saved in a SQLite database.
  This allowed poll workers to record that a registered voter had voted. And the vote had been placed; the database was just correctly updated by the system of whether they voted.
- Verification SMS: It tested the verification SMS sent to voters in relation to voting. The SMS notifications were being sent without any hiccups to the registered phones using the Twilio API.
  Response Handling: Verified that voters receiving the SMS to verify their responses can submit them correctly into the system. For unauthorized voting answers, we marked and input them into the database.

Impact on Voter Confidence

A Streamlit app similar to this could lead the way in restoring some voter confidence back to our political processes. The system addresses significant cases of voting impersonation by verifying each vote using SMS alerts to ensure that only registered voters are able to cast ballots. This would enhance not only the entire electoral system but also electronic voting machines (EVMs) by rendering them more credible.

Limitations

Even though the application is really powerful, it comes with some limitations:

- Data Dependency: The system allows dependencies on voter data to be saved accurately at the campaign end. Any haphazard public database would introduce inconsistencies or out-of-date information that could compromise the accuracy of the application.
- SMS delivery: text this on the reliable SMS delivery is a must verification process. There may be network problems or mismatches in cellphone numbers that mean the verification messages never arrive or take a long time to. The system expects a reply the soonest from voters over SMS verification. In reality, some voters may opt to disregard or end up missing the communications, which in turn can produce cavities inside the verification course of action.

VI. Conclusion

So this study introduces a novel approach to preventingvoting fraud and increasing voter confidence by using a web application via edge technology (Streamlit). The most basic feature of the technology, combined with voter ID, Aadhaar numbers, and cellphone numbers, will ensure that only eligible voters can cast their vote. The program sends post-voting validation messages using the Twilio SMS API and stores voter information in a secure SQLite database. Some of the main findings include: responsible and practical working systems for tallying the vote. Effectiveness at Identifying and Remedying Vote Impersonation Utilization of the SMS Post-election Verification System It is evident from the feedback of users and interested parties that there is much more openness and trust in the political process.

According to the research, this application can greatly lower the number of voter impersonations, which will improve the legitimacy and integrity of elections held in India with electronic voting machines (EVMs).

VII. Future work

Even though the current application yields encouraging results, further research and improvement are needed in a number of areas:

Integration with Government Databases: A future iteration may involve direct integration with government databases, leveraging interoperable data standards, to allow for real-time voter data updating. Here, the updating of missing data like address or contact number will be done from the database of the state government (or any other agency), thus eliminating manual intervention. Advanced Security Measures: Considering the sensitivity of the matter, certain security measures have been considered, such as end-to-end encrypted communication, multi-modal authentication mechanisms, biometric verification, and hash functions.

Multilingual Handle: Making the application available in multiple languages would help ensure broader accessibility to different kinds of communities in different places and in our Indian context across the different language speakers, which fills the criteria of most people in India.

Multiple/Other Verification Measures: To add another layer of assurance, research should be done into the viability of adding alternative verification methods such as email or phone notifications, notifications from mobile apps, biometrics, etc. to curb any potential misuse of the verification systems. Scalability and Performance _ A Study to Check Database Capacity and Performance—Scale the Performance: The next step for this system is to do extensive research and improve the application to maintain big data and user counts so the system performs with the requested response times without faults when the aggregate vote increases. User Experience Updates Maintaining the look and feel of the application through the upgrading process will make it easier for non-tech-savvy individuals to utilize the app.

Solves these problems and provides a way to turn them into a comprehensive and reliable voter confidence and credibility booster for India's election system. A new milestone in secure electronic voting systems will be achieved, enabling future technological advancements, thanks to the basis this research lays the foundation for.

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